

Supplementary Material: Contralateral-Normalized Deformation and Stationary Log-Velocity Decomposition after Chronic Stroke

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1 Complete predictive-model results

Table S1 reports every evaluated and sensitivity feature family. MAE intervals were obtained by resampling participants after averaging each participant’s absolute error across 20 complete outer-CV repetitions. Intervals across repeated splits are not treated as inferential confidence intervals.

Table 1: Performance of all prediction models on the common 210-participant cohort.

Model	MAE (95% CI)	RMSE	R^2	Pearson r
Intercept only	24.14 (22.28, 25.97)	27.68	-0.009	-0.119
Clinical	24.23 (22.36, 25.99)	27.83	-0.020	-0.132
Conventional lesion	16.42 (14.87, 18.09)	20.36	0.454	0.675
Lesion + deformation	16.04 (14.46, 17.72)	20.33	0.456	0.677
Deformation + registration QC	16.00 (14.33, 17.68)	20.40	0.452	0.674
Lesion + uncertainty	15.71 (14.14, 17.36)	20.11	0.468	0.688
Uncertainty + deformation	15.77 (14.23, 17.46)	19.99	0.473	0.689
Lesion + log-velocity Hodge	16.48 (14.88, 18.15)	20.50	0.447	0.670
Deformation + log-velocity Hodge	16.12 (14.57, 17.79)	20.35	0.454	0.676

The direct incremental mean MAE advantage of deformation after lesion uncertainty was -0.06 points (95% participant-bootstrap interval -0.77 to 0.61). The direct incremental advantage of log-velocity Hodge features after displacement features was -0.08 points (-0.35 to 0.21).

2 Complete descriptive-association family

The 12 correlations in Table S2 formed one descriptive multiplicity family. Intervals are participant-bootstrap 95% intervals; p values are Holm adjusted across all rows. These are bivariate descriptions, not covariate-adjusted effects or causal estimates.

3 Log-domain numerical audit

The same numerical construction was used for all 214 cases: a 4-mm analysis grid, a raised-cosine taper over 4 voxels from the valid-domain boundary, Gaussian smoothing with $\sigma = 2.5$ voxels, and 6-voxel zero padding. The stationary logarithm used six scaling-and-squaring steps, no more than six symmetric residual updates, and a maximum relative reconstruction RMSE of 0.02. All 214 cases passed. In the analysis cohort, the median minimum input Jacobian was 0.761 (IQR 0.695–0.806), and median relative exponentiation RMSE was 0.011 (IQR 0.010–0.015).

The Fourier projection was evaluated on the entire padded grid. Nonzero frequencies were projected parallel and perpendicular to their wave vector; the zero-frequency coefficient was retained as the harmonic

Table 2: Descriptive Spearman correlations.

Association	n	Spearman ρ	95% CI	Holm p
Lesion volume vs AQ	210	-0.698	(-0.769, -0.612)	< 0.001
Deformation magnitude vs lesion volume	210	0.407	(0.281, 0.528)	< 0.001
Absolute radial deformation vs lesion volume	210	0.432	(0.304, 0.543)	< 0.001
Magnitude vs registration sensitivity	210	0.399	(0.269, 0.517)	< 0.001
Absolute radial vs registration sensitivity	210	0.387	(0.256, 0.501)	< 0.001
Deformation magnitude vs AQ	210	-0.352	(-0.470, -0.222)	< 0.001
Absolute radial deformation vs AQ	210	-0.368	(-0.492, -0.239)	< 0.001
Signed radial deformation vs AQ	210	-0.235	(-0.356, -0.112)	0.002
Log-velocity RMS vs lesion volume	210	-0.021	(-0.167, 0.116)	0.759
Log-velocity RMS vs AQ	210	0.071	(-0.063, 0.206)	0.613
Curl-free energy fraction vs AQ	210	-0.240	(-0.365, -0.104)	0.002
Divergence-free energy fraction vs AQ	210	0.176	(0.038, 0.303)	0.032

component. Component energies reconstructed total energy to floating-point precision in every case. Because Hodge components on bounded domains depend on boundary conditions, the reported fractions apply only to this stated tapered periodic construction.

4 Left-dominant-lesion sensitivity analysis

Excluding the single right-dominant case left $n = 209$. The mean MAE advantage of displacement features was 0.33 points (95% participant-bootstrap interval -0.51 to 1.09), again failing the superiority criterion.

5 Interpretation boundary

The stationary vector field is a group-log registration parameter with displacement units over an arbitrary unit interval. It is not the original time-dependent velocity optimized by LDDMM and is not physical tissue velocity. The curl-free component, divergence, and log-Jacobian are geometric quantities; without a constitutive model, tissue parameters, loads, and mechanical boundary conditions, none is an estimator of pressure.